

MATH RL Foundations

Homework 1

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1 Question 1

1.1 Part a

For each state $i \in \mathcal{S}$, we define the set of possible return times by:

$$R_i = \{n \geq 1 : (P^n)_{ii} > 0\} = \{n \geq 1 : (P_{ii})^n > 0\} \quad (1)$$

The period of state i is defined as:

$$d(i) = \gcd R_i. \quad (2)$$

Since one state has a self-loop:

$$\exists i \in \mathcal{S}, P_{ii} > 0 \Rightarrow d(i) = 1 \quad (3)$$

Now, we need to prove all other states have the same period of 1. Take any other state $j \in \mathcal{S}$. Since the markov chain is irreducible, there is a path from j to i , and also a path from i to j . Therefore:

$$\exists m, n \geq 1; \quad (P^m)_{ji} > 0 \wedge (P^n)_{ij} > 0 \quad (4)$$

So, state j can return to itself by first going from j to i , and then from i back to j . Hence:

$$(P^m)_{ji} \times (P^n)_{ij} = (P^{m+n})_{jj} > 0 \quad (5)$$

But state i has a self-loop. So, we can stay at state i for one extra step before going back to j . Therefore:

$$(P^m)_{ji} \times P_{ii} \times (P^n)_{ij} = (P^{m+n+1})_{jj} > 0 \quad (6)$$

Thus, both $m+n$ and $m+n+1$ are possible return times for state j . Since these are two consecutive integers:

$$d(j) = \gcd R_j = \gcd(m+n, m+n+1) = 1 \quad (7)$$

Therefore, the markov chain is aperiodic.

1.2 Part b

For this, I can design the following markov chain in which at least one of states can return to itself after 2 or 3, or more moves. Since $\gcd\{2, 3, \dots\} = 1$, then the state is aperiodic. In a finite and irreducible markov chain, if a state is aperiodic, then all states are aperiodic as well.

The aperiodic markov chain is shown by Fig 1, and its transition matrix is:

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}. \quad (8)$$

There is no self-loop:

$$\forall i \in \mathcal{S}, P_{ii} = 0 \quad (9)$$

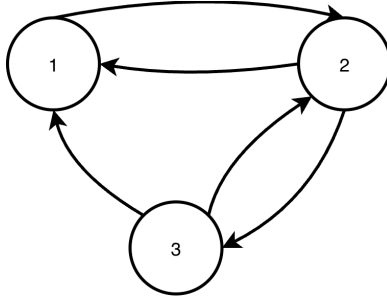


Figure 1: A aperiodic markov chain.

Now, I compute P^2 and P^3 :

$$P^2 = \begin{pmatrix} 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & 0 \end{pmatrix}, \quad P^3 = \begin{pmatrix} 0 & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{8} & 0 & \frac{1}{8} \\ \frac{1}{8} & \frac{1}{8} & 0 \end{pmatrix} \quad (10)$$

So, each state can return to itself in both 2 and 3 steps. And since $\gcd(2, 3) = 1$, that means the the markov chain is aperiodic.

2 Question 2

2.1 Part a

The assumption is that the markov chain has period b . We want to partition State S to $C_0, C_1, \dots, C_{b-1}$ such that:

$$\mathcal{S} = \bigcup_{i=0}^{b-1} C_i \quad (11)$$

$$\forall i, j \in \{0, \dots, b-1\}, \quad i \neq j \implies C_i \cap C_j = \emptyset \quad (12)$$

$$\forall x \in C_i, y \in C_j; \quad P(x, y) > 0 \iff j = (i+1) \% b \quad (13)$$

We can choose one state randomly, call it x_0 , and assume that it belongs to C_0 . Then, we create the other sets based on this state:

$$\forall i \in \{0, \dots, b-1\} \quad C_i = \{x \in \mathcal{S} \mid \exists n \geq 0, (P^n)_{x_0 x} > 0, n \% b = i\} \quad (14)$$

So, $x \in C_i$ means that we can reach x from x_0 after n steps, where $n \% b = i$.

Since the markov chain is irreducible, every state can be reached from x_0 . Therefore, the first condition in Eq 11 is satisfied. Now, we need to prove that every state belongs to only one C_i , Eq 12.

Assume by contradiction that there exists some state x such that

$$\exists i, j \quad x \in C_i \cap C_j \wedge i \neq j \quad (15)$$

Then, there exist $m, n \geq 0$ such that

$$\exists m, n \quad (P^m)_{x_0 x} > 0 \wedge (P^n)_{x_0 x} > 0 \wedge m \% b = i \wedge n \% b = j \quad (16)$$

Since the markov chain is irreducible, we can also go from x back to x_0 . Therefore:

$$\exists r \geq 1 \quad (P^r)_{xx_0} > 0 \quad (17)$$

So, we have two possible return times to x_0 :

$$(P^{m+r})_{x_0x_0} > 0 \wedge (P^{n+r})_{x_0x_0} > 0 \quad (18)$$

Since the period is b , both return times must be divisible by b :

$$(m+r)\%b = 0 \wedge (n+r)\%b = 0 \Rightarrow (n-m)\%b = 0 \quad (19)$$

But by contradictory assumption:

$$m\%b = i \wedge n\%b = j \wedge i \neq j \quad (20)$$

So:

$$(n-m)\%b \neq 0 \quad (21)$$

which is a **contradiction**. Therefore,

$$C_i \cap C_j = \emptyset, \quad i \neq j \quad (22)$$

Now, we have created $C_0, \dots, C_{b-1}$, their union is $\mathcal{S}$, and they are exclusive. Both conditions at equations 11 and 12 is proven.

Now, take some $x \in C_i$, and assume that

$$P_{xy} > 0 \quad (23)$$

Since $x \in C_i$:

$$\exists n \quad (P^n)_{x_0x} > 0 \wedge n\%b = i \quad (24)$$

Then:

$$(P^{n+1})_{x_0y} \geq (P^n)_{x_0x} P_{xy} > 0 \wedge (n+1)\%b = (i+1)\%b \quad (25)$$

Therefore:

$$y \in C_{(i+1)\%b} \quad (26)$$

So, the chain moves as

$$C_0 \rightarrow C_1 \rightarrow \dots \rightarrow C_{b-1} \rightarrow C_0. \quad (27)$$

2.2 Part b

From part a, we know that if $x \in C_i$ and $P_{xy} > 0$, then

$$y \in C_{(i+1)\%b} \quad (28)$$

So, if we order the states based on $C_0, C_1, \dots, C_{b-1}$, the transition matrix P has the following structure:

$$P = \begin{pmatrix} 0 & P_{01} & 0 & \cdots & 0 \\ 0 & 0 & P_{12} & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & P_{b-2,b-1} \\ P_{b-1,0} & 0 & \cdots & 0 & 0 \end{pmatrix} \quad (29)$$

Here, P_{ij} represents the transitions from states in C_i to states in C_j .

We can go **further** with P^b . Starting from some class C_i , after one step we move to $C_{(i+1)\%b}$, and after b steps we move back to C_i . Therefore:

$$x \in C_i \wedge y \in C_j \wedge i \neq j \Rightarrow (P^b)_{xy} = 0. \quad (30)$$

So, P^b has a block diagonal structure:

$$P^b = \begin{pmatrix} Q_0 & 0 & \cdots & 0 \\ 0 & Q_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & Q_{b-1} \end{pmatrix} \quad (31)$$

where:

$$Q_i = (P^b)_{C_i, C_i}. \quad (32)$$

Therefore, the sparsity of P^b is that all transitions between two different classes are zero. The entries inside each Q_i do not necessarily need to be non-zero.

Now, we show that each Q_i is irreducible. Take any two states $x, y \in C_i$. Since the original markov chain is irreducible:

$$\exists n \geq 1 \quad (P^n)_{xy} > 0 \quad (33)$$

Since x and y are both in the same class C_i , the number of steps n must be divisible by b . Therefore:

$$\exists k \geq 1 \quad n = kb \quad (34)$$

Then:

$$(P^{kb})_{xy} = ((P^b)^k)_{xy} = (Q_i^k)_{xy} > 0 \quad (35)$$

Therefore, every state in C_i can reach every other state in C_i using Q_i , so Q_i is irreducible.

Finally, we show that Q_i is aperiodic. Take some state $x \in C_i$. For the original markov chain, define its return times as:

$$R_x = \{n \geq 1 : (P^n)_{xx} > 0\} \quad (36)$$

Since the period of the original markov chain is b :

$$\gcd R_x = b \quad (37)$$

Also, every return time in R_x is divisible by b . So, we can write:

$$R_x = \{bk_1, bk_2, \dots\} \quad (38)$$

For Q_i , one step is equal to b steps in the original markov chain. Therefore, its possible return times are:

$$R_x^{Q_i} = \{k_1, k_2, \dots\} \quad (39)$$

Then:

$$b = \gcd(bk_1, bk_2, \dots) = b \gcd(k_1, k_2, \dots) \quad (40)$$

Therefore:

$$\gcd(k_1, k_2, \dots) = 1 \quad (41)$$

So, Q_i has period 1 and is aperiodic.

Therefore, P^b is block diagonal, and each block Q_i defines an irreducible and aperiodic markov chain on C_i .

3 Question 3

3.1 Part a

The uniform dist is:

$$\pi_u = \left(\frac{1}{|\mathbb{S}|}, \dots, \frac{1}{|\mathbb{S}|} \right) \in \mathbf{R}^{|\mathbb{S}|} \quad (42)$$

We need to prove:

$$\pi_u P = \pi_u \quad (43)$$

So for any j :

$$(\pi_u P)_j = \sum_k (\pi_u)_k \times P_{kj} = \sum_k \frac{1}{|\mathbb{S}|} \times P_{kj} = \frac{1}{|\mathbb{S}|} \times \sum_k P_{kj} = \frac{1}{|\mathbb{S}|} \times 1 = (\pi_u)_j \quad (44)$$

3.2 Part b

The markov chain is *Part a* is a “detailed balance” markov chain:

$$\forall i, j \quad P_{ij} = P_{ji} \quad (45)$$

We also have the relaxed version of detailed balance in which:

$$\forall i, j \quad \sum_i P_{ij} = \sum_j P_{ij} \quad (46)$$

Even though this markov chain is not symmetric, uniform dist. is still its stationary dist. :

$$(\pi_u P)_j = \sum_k (\pi_u)_k \times P_{kj} = \sum_k \frac{1}{|\mathbb{S}|} \times P_{kj} = \frac{1}{|\mathbb{S}|} \times \sum_k P_{kj} = \frac{1}{|\mathbb{S}|} \times \sum_j P_{kj} = \frac{1}{|\mathbb{S}|} \times 1 = (\pi_u)_j \quad (47)$$

So, the relaxed balanced markov chain have uniform stationary dist. An example could be:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \quad (48)$$

4 Question 4

4.1 Part a

We want to prove that:

$$V(x_0) = \lim_{T \rightarrow \infty} V_T(x_0) = \sum_{x \in \mathcal{X}} \pi(x) R(x) \quad (49)$$

We have:

$$V_T(x_0) = \frac{1}{T+1} \mathbb{E} \left[\sum_{t=0}^T R(X_t) \mid X_0 = x_0 \right] \quad (50)$$

Here, t is the time index of the markov chain, while T is the total number of time steps that we move. For a fixed T , we average the rewards from $t = 0$ to $t = T$.

We can write:

$$V_T(x_0) = \frac{1}{T+1} \sum_{t=0}^T \mathbb{E} [R(X_t) \mid X_0 = x_0] \quad (51)$$

For each time t , we know:

$$\mathbb{E} [R(X_t) \mid X_0 = x_0] = \sum_{x \in \mathcal{X}} R(x) \Pr(X_t = x \mid X_0 = x_0) \quad (52)$$

$$= \sum_{x \in \mathcal{X}} R(x) P^t(x_0, x) \quad (53)$$

Now, substituting this into $V_T(x_0)$ gives:

$$V_T(x_0) = \frac{1}{T+1} \sum_{t=0}^T \sum_{x \in \mathcal{X}} R(x) P^t(x_0, x) \quad (54)$$

$$= \sum_{x \in \mathcal{X}} R(x) \left(\frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) \right) \quad (55)$$

Since the markov chain is irreducible and aperiodic, it converges to its unique stationary distribution π . Therefore, for every $x \in \mathcal{X}$:

$$t \rightarrow \infty \implies P^t(x_0, x) \rightarrow \pi(x) \quad (56)$$

Now, we show that:

$$\frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) \rightarrow \pi(x) \quad (57)$$

Since $P^t(x_0, x) \rightarrow \pi(x)$, for every $\epsilon > 0$, there exists some N such that for all $t \geq N$:

$$\forall \epsilon \geq 0, \exists N \geq t, \quad s.t. \quad |P^t(x_0, x) - \pi(x)| < \epsilon \quad (58)$$

Consider:

$$\left| \frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) - \pi(x) \right| \quad (59)$$

$$= \left| \frac{1}{T+1} \sum_{t=0}^T (P^t(x_0, x) - \pi(x)) \right| \quad (60)$$

$$\leq \frac{1}{T+1} \sum_{t=0}^T |P^t(x_0, x) - \pi(x)| \quad (61)$$

We split the sum into two parts:

$$\frac{1}{T+1} \sum_{t=0}^T |P^t(x_0, x) - \pi(x)| \quad (62)$$

$$= \frac{1}{T+1} \sum_{t=0}^{N-1} |P^t(x_0, x) - \pi(x)| + \frac{1}{T+1} \sum_{t=N}^T |P^t(x_0, x) - \pi(x)| \quad (63)$$

The first sum has a fixed number of terms, so:

$$T \rightarrow \infty \Rightarrow \frac{1}{T+1} \sum_{t=0}^{N-1} |P^t(x_0, x) - \pi(x)| \rightarrow 0 \quad (64)$$

For the second sum, since $t \geq N$:

$$\frac{1}{T+1} \sum_{t=N}^T |P^t(x_0, x) - \pi(x)| < \frac{1}{T+1} \sum_{t=N}^T \epsilon \quad (65)$$

$$= \frac{T - N + 1}{T+1} \epsilon \quad (66)$$

$$\leq \epsilon \quad (67)$$

Therefore:

$$\lim_{T \rightarrow \infty} \frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) = \pi(x). \quad (68)$$

So, we have proved:

$$V(x_0) = \lim_{T \rightarrow \infty} V_T(x_0) \quad (69)$$

$$= \sum_{x \in \mathcal{X}} R(x) \lim_{T \rightarrow \infty} \left(\frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) \right) \quad (70)$$

$$= \sum_{x \in \mathcal{X}} \pi(x) R(x) \quad (71)$$

4.2 Part b

Since the N simulations are independent, the random variables $R(Y_1(\epsilon)), \dots, R(Y_N(\epsilon))$ are *I.I.D.*. We can use the following equation:

$$MSE(\theta) = Bias^2(\theta) + Variance(\theta) \quad (72)$$

Hence ($V := V(x_0)$):

$$\mathbb{E}[(\hat{V}_N - V)^2] = \mathbb{V}[\hat{V}_N] + \left(\mathbb{E}[\hat{V}_N] - V\right)^2 \quad (73)$$

Since $|R(x)| \leq R_{\max}$:

$$\mathbb{V}[\hat{V}_N] = \mathbb{V}\left[\frac{1}{N} \sum_{n=1}^N R(Y_n(\epsilon))\right] \quad (74)$$

$$= \frac{1}{N^2} \sum_{n=1}^N \mathbb{V}[R(Y_n(\epsilon))] \quad (75)$$

$$= \frac{1}{N} \mathbb{V}[R(Y_n(\epsilon))] \quad (76)$$

$$= \frac{1}{N} (\mathbb{E}[R(Y_n(\epsilon))^2] - \mathbb{E}[R(Y_n(\epsilon))]^2) \quad (77)$$

$$\leq \frac{1}{N} \mathbb{E}[R(Y_n(\epsilon))^2] \quad (78)$$

$$\leq \frac{1}{N} \mathbb{E}[R_{\max}^2] = \frac{R_{\max}^2}{N} \quad (79)$$

Now, we need to bound the bias. From part a:

$$V = \sum_{x \in \mathcal{X}} \pi(x) R(x) \quad (80)$$

Also, since $Y_n(\epsilon) = X_{n,t_{mix}(\epsilon)}$:

$$\mathbb{E}[\hat{V}_N] = \sum_{x \in \mathcal{X}} P^{t_{mix}(\epsilon)}(x_0, x) R(x) \quad (81)$$

$$\mathbb{E}[\hat{V}_N] = \mathbb{E}\left[\frac{1}{N} \sum_{n=1}^N R(Y_n(\epsilon))\right] \quad (82)$$

$$= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[R(Y_n(\epsilon))] \quad (83)$$

$$= \mathbb{E}[R(Y_n(\epsilon))] \quad (84)$$

$$= \sum_x R(x) \Pr(Y_n^\epsilon = x) \quad (85)$$

$$= \sum_x R(x) \Pr(X_{n,t_{mix}(\epsilon)} = x | X_{n,0} = x_0) \quad (86)$$

Therefore:

$$\left| \mathbb{E}[\widehat{V}_N] - V \right| = \left| \sum_{x \in \mathcal{X}} R(x) (\Pr(X_{n,t_{mix}(\epsilon)} = x | X_{n,0} = x_0) - \pi(x)) \right| \quad (87)$$

$$\leq R_{\max} \sum_{x \in \mathcal{X}} |\Pr(X_{n,t_{mix}(\epsilon)} = x | X_{n,0} = x_0) - \pi(x)| \quad (88)$$

By the definition of $t_{mix}(\epsilon)$:

$$\|\Pr(X_{n,t_{mix}(\epsilon)} = \cdot | X_{n,0} = x_0) - \pi\|_{TV} = \frac{1}{2} \sum_{x \in \mathcal{X}} |P^{t_{mix}(\epsilon)}(x_0, x) - \pi(x)| \leq \epsilon \quad (89)$$

So:

$$\left| \mathbb{E}[\widehat{V}_N] - V \right| \leq 2R_{\max}\epsilon \quad (90)$$

Finally by adding the Bias² and Variance, we have:

$$\mathbb{E} \left[(\widehat{V}_N - V)^2 \right] \leq \frac{R_{\max}^2}{N} + 4R_{\max}^2\epsilon^2 \quad (91)$$

$$= R_{\max}^2 \left(\frac{1}{N} + 4\epsilon^2 \right) \quad (92)$$

5 Question 5

5.1 Part a

We will prove it by **contradictory assumption**.

Contradictory Assumption: Let us assume that x and y are the fixed point for T such that:

$$\exists x, y; \quad x \neq y \wedge T(x) = x \wedge T(y) = y \quad (93)$$

By the α -contraction theorem, we have:

$$\|T(x) - T(y)\| \leq \alpha \|x - y\| \wedge \alpha \in [0, 1) \quad (94)$$

We can expand the left side:

$$\|T(x) - T(y)\| = \|x - y\| \leq \alpha \|x - y\| \quad (95)$$

Since $x \neq y$, we have $\|x - y\| \geq 0$. So, we can divide both side by $\|x - y\|$ without changing the direction of $\leq$:

$$\Rightarrow \frac{\|x - y\|}{\|x - y\|} \leq \alpha \quad (96)$$

$$\Rightarrow 1 \leq \alpha \quad (97)$$

But we already have established that $\alpha \in [0, 1)$. **So, we reached a contradiction and our initial assumption of having two unique fixed points are wrong.**

5.2 Part b

We need to define α -contraction T that would shrink the input while having the fixed point outside of domain.

I define the following T :

$$\mathcal{X} = (0, 2) \quad T(x) = \sqrt{x+2} \quad (98)$$

First, we need to show that $T : \mathcal{X} \rightarrow \mathcal{X}$. For every $x \in (0, 2)$, we have:

$$0 < x < 2 \quad (99)$$

$$\Rightarrow 2 < x+2 < 4 \quad (100)$$

$$\Rightarrow \sqrt{2} < \sqrt{x+2} < 2 \quad (101)$$

$$\Rightarrow T(x) \in (0, 2) = \mathcal{X} \quad (102)$$

Now, we show that T is an α -contraction. For any $x, y \in \mathcal{X}$:

$$|T(x) - T(y)| = |\sqrt{x+2} - \sqrt{y+2}| \quad (103)$$

$$= \frac{|x-y|}{\sqrt{x+2} + \sqrt{y+2}} \quad (104)$$

Since $x, y \in (0, 2)$:

$$\sqrt{x+2} > \sqrt{2} \wedge \sqrt{y+2} > \sqrt{2} \quad (105)$$

$$\Rightarrow \sqrt{x+2} + \sqrt{y+2} > 2\sqrt{2} \quad (106)$$

Hence:

$$|T(x) - T(y)| < \frac{1}{2\sqrt{2}}|x-y| \quad (107)$$

$$\leq \alpha|x-y| \quad (108)$$

Since $\alpha = \frac{1}{2\sqrt{2}} < 1$, T is an α -contraction on $\mathcal{X}$.

Now, we check whether T has a fixed point in $\mathcal{X}$. A fixed point must satisfy:

$$T(x) = x \quad (109)$$

$$\sqrt{x+2} = x \quad (110)$$

$$x+2 = x^2 \quad (111)$$

$$x^2 - x - 2 = 0 \quad (112)$$

$$(x-2)(x+1) = 0 \quad (113)$$

$$\Rightarrow x = 2 \vee x = -1 \quad (114)$$

Since $T(x) = \sqrt{x+2} \geq 0$, $x = -1$ cannot be a fixed point. The only fixed point is $x = 2$; however, $2 \notin \mathcal{X}$.

Therefore, T is an α -contraction on $\mathcal{X} = (0, 2)$, but it has no fixed point in $\mathcal{X}$.

5.3 Part c

We know that $T : \mathcal{X} \rightarrow \mathcal{X}$ is an α -contraction, so:

$$\|T(x) - T(y)\| \leq \alpha\|x - y\|, \quad 0 \leq \alpha < 1 \quad (115)$$

We also assume that $\mathcal{X}$ is a complete metric space. Therefore, if we can construct a Cauchy sequence in $\mathcal{X}$, then that sequence converges to some point that is also in $\mathcal{X}$.

Since $T : \mathcal{X} \rightarrow \mathcal{X}$, all the points $x_0, x_1, x_2, \dots$ belong to $\mathcal{X}$. Now, the distance between two consecutive points is:

$$\|x_{k+1} - x_k\| = \|T(x_k) - T(x_{k-1})\| \quad (116)$$

$$\leq \alpha\|x_k - x_{k-1}\| \quad (117)$$

Applying the same inequality repeatedly:

$$\|x_{k+1} - x_k\| \leq \alpha\|x_k - x_{k-1}\| \quad (118)$$

$$\leq \alpha^2\|x_{k-1} - x_{k-2}\| \quad (119)$$

$$\leq \dots \quad (120)$$

$$\leq \alpha^k\|x_1 - x_0\| \quad (121)$$

Now, for any integers $m > n$. Using the triangle inequality:

$$\|x_m - x_n\| \leq \|x_m - x_{m-1}\| + \|x_{m-1} - x_{m-2}\| + \dots + \|x_{n+1} - x_n\| \quad (122)$$

$$\leq \sum_{k=n}^{m-1} \alpha^k \|x_1 - x_0\| \quad (123)$$

$$= \|x_1 - x_0\| \sum_{k=n}^{m-1} \alpha^k \quad (124)$$

Since $0 \leq \alpha < 1$, we can bound the finite sum by the infinite sum:

$$\|x_m - x_n\| \leq \|x_1 - x_0\| \sum_{k=n}^{m-1} \alpha^k \quad (125)$$

$$\leq \|x_1 - x_0\| \sum_{k=n}^{\infty} \alpha^k \quad (126)$$

$$= \frac{\alpha^n}{1 - \alpha} \|x_1 - x_0\| \quad (127)$$

Since $\alpha < 1$:

$$n \rightarrow \infty \Rightarrow \alpha^n \rightarrow 0 \quad (128)$$

Therefore:

$$m, n \rightarrow \infty \Rightarrow \|x_m - x_n\| \rightarrow 0 \quad (129)$$

So, $x_0, x_1, x_2, \dots$ is a Cauchy sequence.

Since $\mathcal{X}$ is complete, every Cauchy sequence in $\mathcal{X}$ converges to a point that is also in $\mathcal{X}$. Therefore, there exists

some $x^* \in \mathcal{X}$ such that:

$$k \rightarrow \infty \implies x_k \rightarrow x^* \quad (130)$$

This means:

$$k \rightarrow \infty \implies \|x_k - x^*\| \rightarrow 0 \quad (131)$$

Now, we need to show that this limit x^* is a fixed point of T : $T(x^*) = x^*$.

$$\|T(x^*) - x^*\| = \|T(x^*) - T(x_k) + T(x_k) - x^*\| \quad (132)$$

$$\leq \|T(x^*) - T(x_k)\| + \|T(x_k) - x^*\| \quad (133)$$

$$\leq \alpha \|x^* - x_k\| + \|T(x_k) - x^*\| \quad (134)$$

$$= \alpha \|x^* - x_k\| + \|x_{k+1} - x^*\| \quad (135)$$

We already know that

$$k \rightarrow \infty \implies \|x^* - x_k\| \rightarrow 0 \quad (136)$$

Therefore, taking $k \rightarrow \infty$:

$$\|T(x^*) - x^*\| \leq \alpha \cdot 0 + 0 \quad (137)$$

$$= 0 \quad (138)$$

Therefore:

$$T(x^*) = x^* \quad (139)$$

Hence, T has a fixed point in $\mathcal{X}$.

6 Question 6

6.1 Part a

We can prove by **contradiction**.

Contradictory Assumption: There is a norm $\|\cdot\|$ in which $T(x) = Wx$ is α -contractive. Therefore:

$$\exists \|\cdot\| \quad \text{s.t.} \quad \exists \alpha \in [0, 1) \quad \forall x, y \quad \|Wx - Wy\| \leq \alpha \|x - y\| \quad (140)$$

However, if we choose $x = \mathbf{1}$ and $y = 2(\mathbf{1})$:

$$\|Wx - Wy\| = \|W \cdot \mathbf{1} - W \cdot 2(\mathbf{1})\| \quad (141)$$

$$= \|\mathbf{1} - 2(\mathbf{1})\| \quad (142)$$

$$= \|\mathbf{-1}(\mathbf{1})\| \quad (143)$$

$$= \|\mathbf{1}\| \quad (144)$$

$$\not\leq \alpha \|\mathbf{1}\| \quad (145)$$

We have reached to a conclusion that contradicts our initial contradictory assumption. **Therefore, the contradictory assumption is invalid and $T(x) = Wx$ is NOT α -contractive.**

6.2 Part b

The hint from the homework states that if a non-negative irreducible matrix A has spectral radius $r = \rho(A)$, then r is a simple eigenvalue, and the corresponding right and left eigenspaces are one-dimensional.

The matrix W is non-negative and irreducible, and, given the fact that W is stochastic, we know $\rho(W) = 1$. Therefore, the eigenspace associated with the eigenvalue 1 is one-dimensional.

Since W is doubly stochastic, we have $W\mathbf{1} = \mathbf{1}$. Hence, $\mathbf{1}$ is an eigenvector of W associated with the eigenvalue 1. Since the corresponding eigenspace is one-dimensional, every eigenvector associated with the eigenvalue 1 must be a scalar multiple of $\mathbf{1}$.

First, we show that every vector in $\text{Span}(\mathbf{1})$ is a fixed point of W . For any $x \in \text{Span}(\mathbf{1})$, there exists some $a \in \mathbb{R}$ such that $x = a\mathbf{1}$ is the fixed point:

$$Wx = W(a\mathbf{1}) \quad (146)$$

$$= aW\mathbf{1} \quad (147)$$

$$= a\mathbf{1} \quad (148)$$

$$= x \quad (149)$$

Now, we prove that there cannot be a fixed point outside $\text{Span}(\mathbf{1})$ by contradiction.

Contradictory Assumption:

$$\exists y \notin \text{Span}(\mathbf{1}) \quad \text{s.t.} \quad Wy = y \quad (150)$$

Then y is an eigenvector of W associated with the eigenvalue 1. Since $y \notin \text{Span}(\mathbf{1})$, the vectors y and $\mathbf{1}$ are linearly independent. Therefore, the eigenspace associated with eigenvalue 1 would have dimension at least two.

This contradicts the Perron–Frobenius theorem, which states that the eigenspace associated with $\rho(W) = 1$ is one-dimensional.

6.3 Part c

$$\frac{1}{N} \sum_n (Wx)_n = \frac{1}{N} \sum_n \sum_j W_{nj} x_j \quad (151)$$

$$= \frac{1}{N} \sum_j \sum_n W_{nj} x_j \quad (152)$$

$$= \frac{1}{N} \sum_j x_j \sum_n W_{nj} \quad (153)$$

$$= \frac{1}{N} \sum_j x_j \cdot 1 \quad (154)$$

$$= \bar{x} \Rightarrow Wx \in \mathbf{A}(\bar{x}) \quad (155)$$

6.4 Part d

We already know $x \in \mathbf{A}(\bar{x})$, thus:

$$\bar{x} = \frac{1}{N} \sum_n x_n = \frac{1}{N} \mathbf{1}^T x \quad (156)$$

So, now we can rewrite the primary equation as:

$$\|Wx - \bar{x}\mathbf{1}\|_2 = \|Wx - \bar{x}\mathbf{1}\|_2 \quad (157)$$

$$= \|Wx - \left(\frac{1}{N}\mathbf{1}^T x\right)\mathbf{1}\|_2 \quad (158)$$

$$= \|Wx - \mathbf{1}\left(\frac{1}{N}\mathbf{1}^T x\right)\|_2 \quad (159)$$

$$= \left\| \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)x \right\|_2 \quad (160)$$

From part b, we already know that $\text{Span}(\mathbf{1})$ is fixed points for $T(x) = Wx$. So:

$$\bar{x}\mathbf{1} \in \text{Span}(\mathbf{1}) \Rightarrow W \cdot \bar{x}\mathbf{1} = \bar{x}\mathbf{1} \quad (161)$$

Therefore, we can show that:

$$\left\| \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\bar{x}\mathbf{1} \right\|_2 = 0 \quad (162)$$

The proof:

$$\left\| \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\bar{x}\mathbf{1} \right\|_2 = \|W\bar{x}\mathbf{1} - \frac{1}{N}\mathbf{1}\mathbf{1}^T\bar{x}\mathbf{1}\|_2 \quad (163)$$

$$= \|\bar{x}\mathbf{1} - \frac{1}{N}N\bar{x}\mathbf{1}\|_2 = 0 \quad (164)$$

Now, we can go back to Eq 160:

$$\left\| \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)x \right\|_2 = \left\| \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)x - \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\bar{x}\mathbf{1} \right\|_2 \quad (165)$$

$$= \left\| \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)(x - \bar{x}\mathbf{1}) \right\|_2 \quad (166)$$

$$\leq \left\| W - \frac{1}{N}\mathbf{1}\mathbf{1}^T \right\|_2 \|x - \bar{x}\mathbf{1}\|_2 \quad (167)$$

And:

$$\sigma = \left\| W - \frac{1}{N}\mathbf{1}\mathbf{1}^T \right\|_2 \quad (168)$$

Therefore:

$$\|Wx - \bar{x}\mathbf{1}\|_2 \leq \sigma \|x - \bar{x}\mathbf{1}\|_2 \quad (169)$$

Now, from the homework we know $\sigma < 1$.

From part c, we already know that"

$$x \in A(\bar{x}) \Rightarrow Wx \in A(\bar{x}) \quad (170)$$

Therefore, W is a pseudo-contraction on $A(\bar{x})$.

6.5 Part e

It was told that W is symmetric. Therefore, by the spectral theorem, W has real eigenvalues and we can choose its eigenvectors to be **orthonormal**. This means:

$$\forall v_i, v_j \in \text{eigenvector} \quad v_i^T v_j = \sigma_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \quad (171)$$

From the previous parts, we already know that $W\mathbf{1} = \mathbf{1}$; therefore, the first eigenvalue is $\lambda_1 = 1$. The corresponding eigenvector is $\mathbf{1}$. However, since we want the eigenvectors to be orthonormal, we normalize it:

$$v_1 = \frac{1}{\sqrt{N}}\mathbf{1} \quad \Rightarrow \quad \left\| \frac{1}{\sqrt{N}}\mathbf{1} \right\|_2 = 1 \quad (172)$$

Perron–Frobenius theorem. Since W is non-negative, irreducible, and $\rho(W) = 1$, the eigenvalue $\lambda_1 = 1$ is simple.

Also, since $W_{nn} > 0$, every state has a self-loop. Since W is also irreducible, the corresponding Markov chain is aperiodic. Therefore, W is primitive, and all other eigenvalues satisfy:

$$|\lambda_i| < 1, \quad i = 2, \dots, N \quad (173)$$

Because the eigenvalues are orthonormal, it means they are linearly independent, which implies $\text{Span}(V) = \mathbb{R}^N$. Thus, we can write derive $x(0)$ based on the orthonormal eigenvectors:

$$x(0) = \sum_{i=1}^N a_i v_i \quad \wedge \quad a_i = v_i^T x(0) \quad (174)$$

For the first eigenvalue:

$$a_1 = v_1^T x(0) = \frac{1}{\sqrt{N}}\mathbf{1}^T x(0) \quad (175)$$

We have:

$$\bar{x} = \frac{1}{N}\mathbf{1}^T x(0) \quad (176)$$

Therefore:

$$\mathbf{1}^T x(0) = N\bar{x} \quad (177)$$

So:

$$a_1 = \frac{1}{\sqrt{N}}N\bar{x} = \sqrt{N}\bar{x} \quad (178)$$

Therefore:

$$a_1 v_1 = \sqrt{N}\bar{x} \left(\frac{1}{\sqrt{N}}\mathbf{1} \right) = \bar{x}\mathbf{1} \quad (179)$$

we have:

$$x(k) = W^k x(0) \quad (180)$$

$$= W^k \sum_{i=1}^N a_i v_i \quad (181)$$

$$= \sum_{i=1}^N a_i W^k v_i \quad (182)$$

$$= \sum_{i=1}^N a_i \lambda_i^k v_i \quad (183)$$

$$= a_1 \lambda_1^k v_1 + \sum_{i=2}^N a_i \lambda_i^k v_i \quad (184)$$

$$= a_1 v_1 + \sum_{i=2}^N a_i \lambda_i^k v_i \quad (185)$$

$$= \bar{x} \mathbf{1} + \sum_{i=2}^N a_i \lambda_i^k v_i \quad (186)$$

Therefore:

$$x(k) - \bar{x} \mathbf{1} = \sum_{i=2}^N a_i \lambda_i^k v_i \quad (187)$$

Since the eigenvectors are orthonormal:

$$\|x(k) - \bar{x} \mathbf{1}\|_2 = \left\| \sum_{i=2}^N a_i \lambda_i^k v_i \right\|_2 \quad (188)$$

$$\leq \sum_{i=2}^N \|a_i \lambda_i^k v_i\|_2 \quad (189)$$

$$= \sum_{i=2}^N |a_i| |\lambda_i|^k \|v_i\|_2 \quad (190)$$

$$= \sum_{i=2}^N |a_i| |\lambda_i|^k \quad (191)$$

λ_2 is the second largest eigenvalue in absolute value. Thus:

$$\|x(k) - \bar{x} \mathbf{1}\|_2 \leq |\lambda_2|^k \sum_{i=2}^N |a_i| \quad (192)$$

And by considering $A = \sum_{i=2}^N |a_i|$, the problem is solved.

7 Question 7

We have the following gradient descent algorithm with noise:

$$x(k+1) = x(k) - \gamma (\nabla f(x(k)) + e(k)) \wedge \mathbb{E}[e(k)] = 0 \wedge \mathbb{V}[\|e(k)\|_2] = \mathbb{E}[\|e(k)\|_2^2] = \sigma^2 \quad (193)$$

f is μ -convex, the optimal solution x^* is unique, therefore:

$$\nabla f(x^*) = 0 \quad (194)$$

Now, we subtract x^* from both sides of the gradient descent iteration:

$$x(k+1) - x^* = x(k) - x^* - \gamma(\nabla f(x(k)) + e(k)) \quad (195)$$

$$= x(k) - x^* - \gamma \nabla f(x(k)) - \gamma e(k) \quad (196)$$

$$= x(k) - x^* - \gamma(\nabla f(x(k)) - \nabla f(x^*)) - \gamma e(k) \quad (\nabla f(x^*) = 0) \quad (197)$$

Taking the squared Euclidean norm and expectation:

$$\mathbb{E} [\|x(k+1) - x^*\|_2^2] = \mathbb{E} [\|x(k) - x^* - \gamma(\nabla f(x(k)) - \nabla f(x^*))\|_2^2] + \gamma^2 \sigma^2 \quad (198)$$

$$= \mathbb{E} [\|x(k) - x^*\|_2^2] \quad (199)$$

$$- 2\gamma \mathbb{E} [\langle \nabla f(x(k)) - \nabla f(x^*), x(k) - x^* \rangle] \quad (200)$$

$$+ \gamma^2 \mathbb{E} [\|\nabla f(x(k)) - \nabla f(x^*)\|_2^2] + \gamma^2 \sigma^2 \quad (201)$$

Given in the hint:

$$\langle \nabla f(x(k)) - \nabla f(x^*), x(k) - x^* \rangle \geq \frac{L\mu}{L+\mu} \|x(k) - x^*\|_2^2 + \frac{1}{L+\mu} \|\nabla f(x(k)) - \nabla f(x^*)\|_2^2 \quad (202)$$

Therefore:

$$\mathbb{E} [\|x(k+1) - x^*\|_2^2] \leq \left(1 - \frac{2\gamma L\mu}{L+\mu}\right) \mathbb{E} [\|x(k) - x^*\|_2^2] \quad (203)$$

$$+ \left(\gamma^2 - \frac{2\gamma}{L+\mu}\right) \mathbb{E} [\|\nabla f(x(k)) - \nabla f(x^*)\|_2^2] + \gamma^2 \sigma^2 \quad (204)$$

Since:

$$0 < \gamma < \frac{2}{L+\mu} \Rightarrow 0 < \gamma^2 < \frac{2}{L+\mu} \gamma \Rightarrow \gamma^2 - \frac{2\gamma}{L+\mu} < 0 \quad (205)$$

So:

$$\mathbb{E} [\|x(k+1) - x^*\|_2^2] \leq \left(1 - \frac{2\gamma L\mu}{L+\mu}\right) \mathbb{E} [\|x(k) - x^*\|_2^2] + \gamma^2 \sigma^2 \quad (206)$$

We can introduce p :

$$p = 1 - \frac{2\gamma L\mu}{L+\mu} \quad (207)$$

Since:

$$4L\mu \leq (L+\mu)^2 \wedge \gamma < 2/(L+\mu) \Rightarrow 0 < p < 1 \quad (208)$$

Thus:

$$\mathbb{E} [\|x(k+1) - x^*\|_2^2] \leq p \mathbb{E} [\|x(k) - x^*\|_2^2] + \gamma^2 \sigma^2 \quad (209)$$

$$\leq p^{k+1} \mathbb{E} [\|x(0) - x^*\|_2^2] + \gamma^2 \sigma^2 \sum_{i=0}^k p^i \quad (210)$$

$$\leq p^{k+1} \mathbb{E} [\|x(0) - x^*\|_2^2] + \frac{\gamma^2 \sigma^2}{1-p} \quad (211)$$

$$= \left(1 - \frac{2\gamma L \mu}{L + \mu}\right)^{k+1} \mathbb{E} [\|x(0) - x^*\|_2^2] + \gamma \sigma^2 \frac{L + \mu}{2L \mu} \quad (212)$$

Now, we need to answer **what the cost of high solution accuracy is**.

$$\limsup_{k \rightarrow \infty} \mathbb{E} [\|x(k+1) - x^*\|_2^2] \leq \gamma \sigma^2 \frac{L + \mu}{2L \mu} \quad (213)$$

So, by choosing γ sufficiently small, the final error can be made arbitrarily small. However, when γ becomes smaller:

$$1 - \frac{2\gamma L \mu}{L + \mu} \rightarrow 1 \quad (214)$$

which makes the convergence slower.

Therefore, the cost of high solution accuracy is slower convergence and more required iterations.

8 Question 8

We need to give an example of a sequence of random variables that converges in probability, but does not converge almost surely:

$$\{X_1, X_2, \dots\} \wedge \lim_{k \rightarrow \infty} \Pr(|X_k - X^*| \geq \epsilon) = 0 \wedge \Pr\left(\lim_{k \rightarrow \infty} X_k = X^*\right) \neq 1 \quad (215)$$

Let us make the sequence using Bernoulli distribution:

$$X_n \sim \text{Bernoulli}\left(\frac{1}{n}\right) \Rightarrow \Pr(X_n) = \begin{cases} 1 - \frac{1}{n} & X_n = 0 \\ \frac{1}{n} & X_n = 1 \end{cases} \quad (216)$$

First, we prove that X_n converges in probability to $X^* = 0$. For any $\epsilon \in (0, 1]$:

$$\forall \epsilon \geq 1, \exists n \in \mathbb{N}, \Pr(|X_n - 0| \geq \epsilon) = \Pr(|X_n| > 0) \quad (217)$$

$$= \Pr(X_n = 1) \quad (X_n \in \{0, 1\}) \quad (218)$$

$$= \frac{1}{n} \xrightarrow{n \rightarrow \infty} 0 \quad (219)$$

Therefore, X_n converges in probability to 0.

Now, we need to prove that X_n does not converge almost surely to 0. **Since $X_n \in \{0, 1\}$, convergence to 0 would require that after some finite N , all the random variables are equal to 0.**

Now, we prove by **contradiction**.

Contradictory Assumption: Assume that X_n converges almost surely to 0. Then, with probability 1, there exists some finite N such that:

$$\exists N \in \mathbb{N}, \forall n \geq N \quad \text{s.t.} \quad X_n = 0 \quad (220)$$

Hence:

$$\mathbb{P}(X_n = 0, \forall n \geq N) = \prod_{n=N}^{\infty} \left(1 - \frac{1}{n}\right) = \prod_{n=N}^{\infty} \frac{n-1}{n} \quad (221)$$

We can first consider the finite product for any $M > N$:

$$\forall M > N \quad \prod_{n=N}^M \frac{n-1}{n} = \frac{N-1}{N} \frac{N}{N+1} \frac{N+1}{N+2} \cdots \frac{M-1}{M} \quad (222)$$

$$= \frac{N-1}{M} \quad (223)$$

Now, the infinite product in Eq 221 is the limit of this finite product as $M \rightarrow \infty$:

$$\mathbb{P}(X_n = 0, \forall n \geq N) = \prod_{n=N}^{\infty} \frac{n-1}{n} \quad (224)$$

$$= \lim_{M \rightarrow \infty} \prod_{n=N}^M \frac{n-1}{n} \quad (225)$$

$$= \lim_{M \rightarrow \infty} \frac{N-1}{M} \quad (226)$$

$$= 0. \quad (227)$$

We reached a contradiction with our initial assumption.

9 Question 9

9.1 Part a

We want to prove Chebyshev's inequality using Markov's inequality. We define the following non-negative random variable:

$$Y = (X - \mu)^2 \geq 0 \quad (228)$$

By Markov's inequality:

$$\forall \alpha > 0 \quad \Pr(Y \geq \alpha^2) \leq \frac{\mathbb{E}[Y]}{\alpha^2} \quad (229)$$

Now, we substitute $Y = (X - \mu)^2$:

$$\Pr((X - \mu)^2 \geq \alpha^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{\alpha^2} \quad (230)$$

Since:

$$(X - \mu)^2 \geq \alpha^2 \iff |X - \mu| \geq \alpha \quad (231)$$

we have:

$$\Pr(|X - \mu| \geq \alpha) \leq \frac{\mathbb{E}[(X - \mu)^2]}{\alpha^2} = \frac{\mathbb{V}[X]}{\alpha^2} \quad (232)$$

9.2 Part b

We have:

$$S_n = \frac{1}{n} \sum_{i=1}^n X_i \quad (233)$$

First, we compute its mean:

$$\mathbb{E}[S_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \quad (234)$$

$$= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] \quad (235)$$

$$= \frac{1}{n} \sum_{i=1}^n \mu \quad (236)$$

$$= \mu \quad (237)$$

Since $X_1, \dots, X_n$ are *I.I.D.* and each has variance σ^2 :

$$\mathbb{V}[S_n] = \mathbb{V}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \quad (238)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}[X_i] \quad (239)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 \quad (240)$$

$$= \frac{\sigma^2}{n} \quad (241)$$

Now, using Chebyshev's inequality:

$$\Pr(|S_n - \mu| \geq \epsilon) \leq \frac{\mathbb{V}[S_n]}{\epsilon^2} \quad (242)$$

$$= \frac{\sigma^2}{\epsilon^2 n} \xrightarrow{n \rightarrow \infty} 0 \quad (243)$$

9.3 Part c

We want the minimum number of samples that satisfies the requirement of the question. Therefore, the answer is:

$$\inf\{n \in \mathbb{N}, n \geq 1 \mid \Pr(|S_n - \mu| \leq \epsilon) \geq 1 - \delta\} \quad (244)$$

From part b:

$$\Pr(|S_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2 n} \quad (245)$$

So, we can rewrite $\Pr(|S_n - \mu| \leq \epsilon)$:

$$\Rightarrow \Pr(|S_n - \mu| \leq \epsilon) \geq 1 - \delta \quad (246)$$

$$\Rightarrow 1 - \Pr(|S_n - \mu| \leq \epsilon) \leq 1 - (1 - \delta) \quad (247)$$

$$\Rightarrow \Pr(|S_n - \mu| \geq \epsilon) \leq \delta \quad (\text{Using } \Pr(|S_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2 n}) \quad (248)$$

$$\Rightarrow \frac{\sigma^2}{\epsilon^2 n} \leq \delta \quad (249)$$

$$\Rightarrow n \geq \frac{\sigma^2}{\epsilon^2 \delta} \quad (250)$$

Thus, the final answer is:

$$\inf \left\{ n \in \mathbb{N}, n \geq 1 \mid n \geq \frac{\sigma^2}{\epsilon^2 \delta} \right\} \quad (251)$$